

Prompt 2: AI Metrics and Data Quality — Presentation Guide

Full prompt: *“What metrics are important for evaluating the impact of AI on processing and data management workflows, and how does data quality play a role?”*

Scenario: Teacher–Student roleplay for Bloomberg Data AI Trainer interview. You present as the teacher; interviewers play trainees. 20 min presentation within a 60 min session.

Table of Contents

1. Overview & Build
 2. Presentation Structure — 8-phase chronological flow
 3. Slide-by-Slide Content & Talking Points — by phase
 4. Key Concepts Reference
 5. Worked Examples Summary
 6. References & Links
-

Overview & Build

Output: presentation.pdf

./build.sh

Requires `pdflatex` (LaTeX distribution).

Presentation Structure

Slides follow a chronological flow: **Setup** → **Foundation** → **Taxonomy** → **AI Metrics** → **Governance & Operations** → **Execution** → **Practice** → **Wrap-up**.

Phase	Slide	Purpose
1. Setup	Title	Set context
	The Prompt	Display exact question; opening scenario
	Purpose	Frame scenario, format
	Why This Matters	Client impact (trust, breaks, time-to-insight)

Phase	Slide	Purpose
2. Foundation	Bloomberg Trust Frame	Core framing
	Where AI Touches Bloomberg	LLM vs Traditional ML use cases; DL+ schema mapping
	Data Quality Is the Constraint	Core principle
	Data Quality Dimensions	Accuracy, completeness, timeliness, consistency, validity
3. Taxonomy	Five-Block Taxonomy: Overview	Five blocks, five decisions
	Five-Block: Data Quality & Operational	Blocks 1 & 2
	Five-Block: Human, AI-Specific & Governance	Blocks 3, 4 & 5
	How to Choose Which Block First	Decision flow
4. AI Metrics	LLM vs Traditional ML	Metrics by AI type
	LLM Evaluation (RAGAS, LLM-as-judge, human-in-the-loop, Pydantic)	Faithfulness, answer relevancy, schema validation
	MLflow GenAI & Component-Aware Governance Metrics	Separate retrieval from generation
		Auditability, traceability, compliance, safety
5. Governance & Operations	Processing & Data Management Metrics	Throughput, latency, freshness
	Measurement Playbook	Measure, gate, ship
	Error Budget & Statistical	SLO, budget, practical significance
	Champion–Challenger Implementation	6 steps
7. Practice	Worked Example: Event-Driven Feeds LLM	LLM scenario
	Worked Example: Document Search RAG	RAG scenario

Phase	Slide	Purpose
8. Wrap-up	Worked Example: Anomaly Detection	Traditional ML scenario
	Tests-as-Code	Great Expectations, dbt, Airflow
	Interactive Moment	Ship vs guardrail question
	Lab Blueprint & Summary	Takeaways
	References	Links

Slide-by-Slide Content & Talking Points

Flow logic: Start with context (Setup), establish why data quality matters (Foundation), introduce the taxonomy (Taxonomy), then show how to measure AI specifically (AI Metrics). Cover governance and operational metrics (Governance & Operations), then how to run experiments (Execution). Apply concepts to real scenarios (Practice), and close with tools and takeaways (Wrap-up).

Phase 1: Setup

Title

Content: Prompt 2: AI Metrics and Data Quality — Metric-Driven Evaluation of AI in Bloomberg Data Workflows

Talking points: Introduce yourself; state the topic.

Timing: 30 sec

The Prompt

Content: The exact prompt in a block: > *“What metrics are important for evaluating the impact of AI on processing and data management workflows, and how does data quality play a role?”*

Opening scenario (30 sec): *“A team ships an AI change that speeds up ingestion by 2×. Clients report wrong data. What went wrong? We’ll answer that today.”*

Visual suggestion: Simple “before/after” or “problem” graphic: “2× faster” with a checkmark, “Wrong data” with an X—or a pipeline icon with a crack/warning to suggest “speed without quality = failure.”

Talking points: This is the question we’re answering. Every slide connects back to it. The scenario hooks attention—faster isn’t better if quality degrades.

Timing: 30 sec

Activity (choose one): - **Mural:** Create a board with the prompt at the centre. Ask trainees to add sticky notes: “One metric you’d care about for an AI change in your workflow” and “One way data quality could go wrong.” Cluster as you discuss. Link: mural.co

Purpose

Content: - **Scenario:** Teacher–Student roleplay for Bloomberg Data AI Trainer interview. You present as the teacher; interviewers play trainees. - **Purpose:** Train Data teams on metrics for evaluating AI impact and the role of data quality; equip trainees to make safe, client-trust-preserving decisions when shipping AI changes. - **Format:** 20 min teach-back; interviewers as trainees. Expect questions mid-presentation.

Talking points: Set expectations. This is training, not a lecture—trainees will engage.

Timing: 30 sec

Why This Matters (Client Impact)

Content: - **Client trust:** Wrong or stale data erodes confidence; metrics help us ship changes that preserve or improve quality - **Fewer breaks, faster resolution:** Reconciliation and anomaly detection metrics reduce ops burden and client-facing incidents - **Faster time-to-insight:** Document Search and Event-Driven Feeds deliver value when answers are faithful and timely—metrics ensure we don’t regress

Visual suggestion: Three icons or simple illustrations: (1) handshake or shield for “trust”; (2) broken link → fixed link for “fewer breaks”; (3) clock or lightning for “faster insight.” Or a simple three-column layout with one headline per column.

Talking points: Connect metrics to outcomes. This isn’t academic—it’s about client trust and operational reliability.

Timing: 45 sec

Bloomberg Trust Frame

Content: - Bloomberg’s AI = “actionable, trustworthy insights” — measurement must include quality, risk, auditability, not just speed or cost - **Impact** =

measurable improvement *without degrading data quality* - **Foundation:** Transparency, Control, Quality & Stability

Talking points: This is the lens. Operational wins are net negative if quality degrades. Client trust depends on data remaining fit for use.

Timing: 1.5 min

Presenter notes on Foundation breakdown: see prompt-2.md.

Where AI Touches Bloomberg

Content: - **LLM / GenAI:** Document Search (400M+ docs, RAG, AI summaries); Event-Driven Feeds (10K+ stories/day, NSTM) - **Traditional ML:** Anomaly detection (ingestion, pricing); GICS/ESG classification; Reconciliation (post-trade); DL+ schema mapping (client→FIGI/CUSIP) - **Scale:** Data License 50M securities, 56K fields; B-PIPE 35M instruments, 330+ exchanges - **Processing & Data Management:** Ingestion (B-PIPE, Kafka), transformation, validation—AI touches these pipeline stages; metrics must cover throughput, latency, freshness.

Talking points: Anchor to Bloomberg. These are real surfaces where metrics matter.

Timing: 1 min

Expanded details & examples:

LLM / GenAI - Document Search & Analysis: 400M+ documents (filings, transcripts, research, news). Users ask in natural language; RAG retrieves relevant chunks and the LLM generates grounded answers with citations. - *Examples:* - “What were the key risks in Apple’s 10-K?” → AI summarises and cites sections - “Show me ESG controversies for this issuer” → grounded answer with source attribution - Metrics: faithfulness, citation correctness. - **Event-Driven Feeds:** 10K+ news stories daily. NSTM (Key News Themes) uses AI to cluster by topic and sentiment. - *Examples:* - Tag incoming story by issuer (e.g., AAPL), sector (Technology), sentiment (positive/negative) - Classify story for portfolio managers and algo trading - Metrics: classification precision/recall, timeliness.

Traditional ML - Anomaly detection: ML task: Unsupervised / one-class classification. Flags outliers in ingestion or pricing. - *Examples:* - Sudden price spike vs. history - Missing fields in a normally complete feed - Volume anomalies - Metrics: precision, recall, time to detect. - **Reconciliation:** ML task: Classification or matching. Compares execution vs. settlement; flags breaks. - *Examples:* - Trade executed at 100.50, settlement shows 100.52 → break flagged for ops - Match execution record to settlement record; classify as “match” or “break” - Metrics: break detection rate, time to resolve. - **DL+ (Data License Plus) schema mapping:** Suggests client field → FIGI/CUSIP mappings; re-

duces manual mapping effort for enterprise data teams. ML/rule-based hybrid. - *What it is*: Client has internal field names; DL+ suggests mappings to Bloomberg identifiers (FIGI, CUSIP). AI assists with suggestive mapping. - *Metrics*: Mapping accuracy (correct suggestions / total); time to first mapping; human override rate (how often users reject suggestions)

B-PIPE (Bloomberg Market Data Feed): Real-time market data feed. **Processing pipelines**: ingestion (B-PIPE Feed Handler, Kafka-based meta-data ingestion), transformation, validation. AI/ML can assist with anomaly detection or predictive maintenance on these pipelines. Used by algo trading, risk systems; 35M instruments, 330+ exchanges.

Phase 2: Foundation

Data Quality Is the Constraint

Content: - **Analogy**: Data quality is like the foundation of a building—AI can't fix cracks in the foundation - Operational wins only matter if data stays fit for client use - “Measure, gate, ship” vs “Ship and hope” - Throughput and automation rate are vanity metrics if accuracy or completeness degrades - Every AI change must answer: *“Did we improve or maintain quality while improving efficiency?”*

Visual suggestion: Building foundation analogy—simple sketch of a building with a cracked foundation (labelled “Data quality”) and an AI/system layer on top. Caption: “AI can't fix cracks in the foundation.” Or a layered diagram: Data (bottom) → Processing → AI (top).

Talking points: Core principle. Use the analogy: AI builds on data; bad data = bad outputs. A pipeline 2× faster that degrades accuracy is a loss.

Timing: 1.5 min

Data Quality Dimensions

Content: Accuracy, Completeness, Timeliness, Consistency, Validity — with impact on AI for each (Wang & Strong, ISO 25012, DAMA)

Talking points: Brief. Wrong inputs → wrong predictions; gaps cause model failure; stale data degrades real-time AI.

Timing: 45 sec

Phase 3: Taxonomy

Five-Block Taxonomy — Overview

Content: Five blocks, five decisions. Each block answers one question and leads to one decision.

Source: Decision-oriented metrics taxonomy derived from Wang & Strong (1996, data quality dimensions), ISO 25012, DAMA, and DORA (operational workflow metrics). Synthesized for Bloomberg Data AI Trainer in prompt-a-deep-research-report.md.

Block	The question you're asking	Your decision	In one line
1. Data quality	Is the data itself correct enough to ship?	Ship / hold	"Can we trust the inputs?"
2. Operational workflow	Is the pipeline running reliably?	Deploy / rollback	"Does the system stay up?"
3. Human workflow	Did we reduce manual work without making things worse?	Automate / keep manual	"Did we save time safely?"
4. AI-specific	Did the AI step add value without introducing unreliability?	Promote / reject	"Is the AI helping or hurting?"
5. Governance & risk	Can we prove what happened and why?	Audit / remediate	"Can we explain and defend it?"

Visual suggestion: Flow diagram with 5 boxes in a row or column, each labelled with the block name and its decision (Ship/hold, Deploy/rollback, etc.). Arrows showing the decision flow. Optional: colour-code Blocks 1–2 as “foundation” (e.g., grey) and Blocks 3–5 as “when AI is in the path” (e.g., blue).

Talking points: Each block = one question, one decision. Blocks 1 and 2 are foundations—fix these before worrying about AI impact.

Timing: 1 min

Five-Block — Block 1 & 2: Data Quality & Operational

Content:

Block 1 — Data quality

“Is the data itself correct enough to ship?” - **What it means:** Are the inputs accurate, complete, timely, consistent? Wrong or stale data → wrong outputs, no matter how good the AI is. - **When to use:** Before any AI evaluation. If data is bad, fix data first. - **Metrics:** Accuracy, completeness, timeliness, consistency, validity - **Bloomberg example:** Data License fields, Event-Driven Feeds story metadata—are we shipping correct, complete data?

Block 2 — Operational workflow

“Is the pipeline running reliably?” - **What it means:** Does the system deploy cleanly, recover from failures, avoid constant breakage? - **When to use:** Before measuring AI impact. You can’t tell if AI is helping if the pipeline is flaky. - **Metrics:** Lead time, change failure rate, MTTR (mean time to repair) - **Bloomberg example:** B-PIPE ingestion, Kafka pipelines—are they stable enough to evaluate changes?

Talking points: Block 1 = trust the inputs. Block 2 = trust the system. Both are prerequisites.

Timing: 1 min

Five-Block — Block 3, 4 & 5: Human, AI-Specific & Governance

Content:

Block 3 — Human workflow

“Did we reduce manual work without making things worse?” - **What it means:** If we automated something (e.g., Document Search summaries), did we actually save hours? Did error rate go up or down? - **When to use:** When AI replaces or assists human tasks—summarization, classification review, search. - **Metrics:** Hours saved, error rate delta - **Bloomberg example:** Document Search—do analysts spend less time and make fewer mistakes?

Block 4 — AI-specific

“Did the AI step add value without introducing unreliability?” - **What it means:** Is the AI output good enough? Faithful, relevant, accurate—or is it hallucinating, off-topic, wrong? - **When to use:** Whenever there is an AI/ML step in the path—RAG, classification, anomaly detection. - **Metrics:** LLM: RAGAS, LLM-as-judge, human-in-the-loop, Pydantic, faithfulness; Traditional: precision, recall, F1 - **Bloomberg example:** Event-Driven Feeds LLM, NSTM (Key News Themes), anomaly detection on ingestion

Block 5 — Governance & risk

“Can we prove what happened and why?” - **What it means:** Can we trace outputs to sources? Log model/prompt versions? Show regulators or clients what we did? - **When to use:** For all AI products—auditability and traceability are always required. - **Metrics:** Auditability, traceability, compliance,

safety policy triggers - **Bloomberg example:** Document Search must preserve attribution; all AI products need decision records

Talking points: Block 4 is the core AI question: value without unreliability. Block 5 is the “can we defend this?” question.

Timing: 1.5 min

How to Choose Which Block First

Content: Decision flow—answer in order:

1. **Is the pipeline healthy?** (Broken deployments? Constant rollbacks? High MTTR?)
→ **No** → Block 2 (Operational) first. Fix the system before evaluating AI.
2. **Is the data correct enough?** (Wrong values? Missing fields? Stale?)
→ **No** → Block 1 (Data quality) first. AI can’t fix bad data.
3. **Did we add or change an AI step?**
→ **Yes** → Block 4 (AI-specific) + Block 5 (Governance). Measure AI impact and ensure we can audit it.
4. **Did we remove manual work?**
→ **Yes** → Block 3 (Human workflow). Track hours saved and error rate.

Visual suggestion: Decision tree or flowchart: Start → “Pipeline healthy?” → No → Block 2; Yes → “Data correct?” → No → Block 1; Yes → “AI step added?” → Yes → Blocks 4+5; “Manual work removed?” → Yes → Block 3.

Talking points: You can’t evaluate AI on a broken pipeline. AI can’t fix bad data.

Timing: 1 min

Phase 4: AI Metrics

LLM vs Traditional ML: When to Use Which Metrics

Content: - **LLM / GenAI:** Faithfulness, answer relevancy, context recall/precision — Document Search, Event-Driven Feeds, NSTM - **Traditional ML:** Precision, recall, F1, time-to-detect — Anomaly detection, GICS/ESG classification, Reconciliation, DL+ schema mapping - Both depend on data quality

Expanded guidance:

Use LLM metrics when: The output is free-form text (summaries, answers, classifications from an LLM). You care whether the model hallucinates, stays

on-topic, and cites sources correctly. - *Document Search (400M docs)*: Faithfulness (claims supported by context); answer relevancy; citation correctness - *Event-Driven Feeds / NSTM (10K+ stories/day)*: Classification precision/recall; timeliness; structured metadata correctness - *Tools*: RAGAS (RAG-focused), MLflow GenAI (Bloomberg Labs stack), LLM-as-a-judge (subjective scoring), human-in-the-loop (human review/validation), Pydantic (structured output validation)

LLM evaluation toolkit (Bloomberg-relevant): - **RAGAS**: Strong for RAG pipelines—faithfulness, answer relevancy, context recall/precision. Good for Document Search and any retrieval-grounded QA. Automated, no human labels needed. - **LLM-as-a-judge**: Use a second LLM to score outputs (e.g., “Rate faithfulness 1–5”, “Is this summary accurate?”). MLflow GenAI supports custom LLM judges. Useful when ground truth is expensive or subjective—e.g., NSTM theme quality, summarization tone. - **Human-in-the-loop**: Human reviewers validate, correct, or approve AI outputs. Use for high-stakes decisions (e.g., promotion), edge cases, or when LLM-as-a-judge is insufficient. Human eval = ground truth for calibration; sample production traffic for ongoing quality. - **Pydantic**: Validate LLM outputs against a schema. For Event-Driven Feeds and NSTM, outputs often must be structured (issuer, sector, sentiment, entity IDs). Validation success rate = metric; parse failures = quality signal. Fits Bloomberg’s need for machine-readable, schema-bound metadata. - **MLflow GenAI**: Bloomberg Labs evaluation harness—built-in scorers (answer_relevance, toxicity), custom scorers, LLM judges, trace evaluation. Versioned evaluation sets; compare prompt/model variants; promote only when guardrails hold.

Use Traditional ML metrics when: The output is structured (labels, matches, binary decisions). You care about correct vs incorrect predictions. - *Anomaly detection*: Precision (few false alarms), recall (catch real anomalies), time to detect - *GICS/ESG classification*: Classification accuracy, inter-rater agreement - *Reconciliation*: Break detection rate, false break rate, time to resolve

Rule of thumb: If it generates or interprets text → LLM metrics. If it predicts a label or match → precision, recall, F1.

Talking points: Choose metrics by AI type. For LLM: RAGAS for RAG; LLM-as-a-judge for subjective scoring; human-in-the-loop for high-stakes validation; Pydantic for structured outputs (Event-Driven Feeds, NSTM); MLflow GenAI as the evaluation harness. For traditional ML: precision/recall.

Timing: 1.5 min

LLM Evaluation: RAGAS, LLM-as-Judge, Pydantic

Content: - **RAGAS:** Faithfulness, answer relevancy, context recall/precision — automated, RAG-focused; good for Document Search - **LLM-as-a-judge:** Second LLM scores outputs (e.g., “Rate faithfulness 1–5”); MLflow GenAI supports custom judges; useful for subjective metrics (NSTM theme quality, summarization) - **Human-in-the-loop:** Human reviewers validate or approve AI outputs; ground truth for promotion decisions; sample production for ongoing quality - **Pydantic:** Schema validation for structured outputs; validation success rate = metric; parse failures = quality signal. Event-Driven Feeds, NSTM metadata (issuer, sector, sentiment) must conform to schema - **RAGAS metrics:** Faithfulness (0–1; <0.8 investigate, 0.9+ acceptable); Answer relevancy; Context recall (retrieval); Context precision (noise)

Talking points: RAGAS for RAG; LLM-as-judge when ground truth is expensive; human-in-the-loop for promotion and high-stakes validation; Pydantic for schema-bound outputs (Bloomberg’s Event-Driven Feeds, NSTM).

Timing: 1.5 min

MLflow GenAI & Component-Aware Evaluation

Content: - MLflow: built-in scorers, `mlflow.genai.evaluate()`, per-record and aggregate scores - **Component-aware:** Separate retrieval from generation. Regressions can come from: Data, Retrieval, Prompts, Model - Retrieval regression → context recall/precision drop. Generation regression → faithfulness/relevancy drop with retrieval stable

Visual suggestion: Four-box diagram: Data → Retrieval → Prompts → Model, with arrows between them. Label each with “If regression here → [metric] drops.” Or a simple RAG flow: User query → Retrieval (context recall/precision) → Generation (faithfulness/relevancy) → Answer.

Talking points: Critical: you must separate retrieval from generation to fix the right thing.

Timing: 1 min

Phase 5: Governance & Operations

Governance Metrics

Content: Auditability, Traceability, Compliance, Safety policy triggers (OWASP LLM Top 10)

Talking points: Can we prove what happened? Document Search must preserve attribution.

Timing: 45 sec

Processing & Data Management Metrics

Content: - **Processing** = ingestion + transformation + validation—the stages where AI touches data workflows. The prompt asks about metrics for both processing and data management. - **Processing:** Ingestion (throughput, latency), transformation (accuracy, error rate), validation (pass rate, freshness); automation rate - **Data management:** Ingestion rate, transformation accuracy, data freshness, storage efficiency - AI depends on timely data

Visual suggestion: Pipeline diagram: three boxes in sequence—Ingestion → Transformation → Validation. Under each, list key metrics (e.g., under Ingestion: throughput, latency). Optional: show AI touching one or more stages with a dotted “AI assist” label.

Talking points: AI often gates or accelerates ingestion. Track freshness.

Timing: 45 sec

Phase 6: Execution

Measurement Playbook

Content: “Measure, gate, ship” — six steps: 1. Define SLIs and SLOs 2. Derive error budget 3. Champion–challenger 4. Canary 5. Rollback triggers 6. Decide upfront what success means (primary, guardrail, rollback)

Talking points: Not “ship, then hope.” Define success before you ship.

Timing: 1.5 min

Error Budget & Statistical Considerations

Content: - Error budget example: 99.9% SLO → 0.1% budget; 1M requests → 1,000 failures allowed; pause at 80% consumed - Practical significance alongside statistical significance; sequential testing, not p-value peeking

Talking points: Define minimum effect size you care about.

Timing: 45 sec

Champion–Challenger Implementation

Content: Six steps: Define slice; Run in parallel; Compare; Promotion criteria; Rollback plan; Duration

Talking points: Same input to champion and challenger; promote only when primary baseline and guardrails hold.

Timing: 45 sec

Phase 7: Practice

Worked Example — Event-Driven Feeds LLM

Content: - **Scenario:** LLM-assisted topic classification for 10K+ daily stories; preserve attribution - **Baseline:** Precision/recall >0.9/0.85; timeliness <60s (p95); failure rate <0.1% - **AI evaluation:** RAGAS, LLM-as-judge, human-in-the-loop, Pydantic, MLflow GenAI; fixed, versioned evaluation set - **Experiment:** Champion–challenger; canary; rollback on faithfulness drop

Talking points: Concrete Bloomberg example. Metrics, experiment design, rollback.

Timing: 2 min

Worked Example — Document Search RAG

Content: - **Scenario:** RAG over 400M+ documents; grounded answers with citations - **Metrics:** Context recall >0.85; Faithfulness >0.9; Citation correctness 100%; Latency (p95) <3s - **Component-aware:** Retrieval vs generation regression - Champion–challenger: keyword + manual vs RAG

Talking points: RAG-specific. Component-aware: retrieval drop vs generation drop.

Timing: 1 min

Worked Example — Anomaly Detection (Traditional ML)

Content: - **Scenario:** AI-assisted anomaly detection for ingestion/pricing; reduce false positives - **Metrics:** Precision >0.85; Recall >0.9; Time to detect <5 min; Alert fatigue <20% - **Data quality dependency:** Historical distribution quality; baseline stability - **Experiment:** Champion–challenger vs rule-based; rollback if precision <0.8 or recall <0.85

Talking points: Traditional ML example. Precision/recall, not faithfulness.

Timing: 1 min

Phase 8: Wrap-up

Tests-as-Code

Content: Great Expectations, dbt, Airflow. Every AI change ships with executable quality checks, publishable evidence, monitored operations.

Talking points: Quality metrics must be executable and enforced at change-time.

Timing: 45 sec

Interactive Moment

Content: Question: “Which metric would you trust to decide if we ship this AI change, and which metric is a guardrail?” - **Ship metric:** Faithfulness; classification precision - **Guardrail:** Error rate; timeliness; compliance adherence

Talking points: Pause for discussion. This demonstrates the core skill.

Timing: 1–2 min

Lab Blueprint & Summary

Content: - **Labs:** A (quality gate), B (AI impact evaluation), C (responsible AI tools) - **Key takeaways:** Data quality is constraint; five-block taxonomy; LLM vs Traditional ML; component-aware; measure-gate-ship; observability & governance prerequisite

Talking points: Recap. Observability and governance are prerequisites for measuring AI impact reliably.

Timing: 1 min

References

Content: Links to Bloomberg Data, Bloomberg AI, Event-Driven Feeds, Data AI Trainer Job, MLflow GenAI, RAGAS, Pydantic, Great Expectations, OWASP LLM Top 10, NIST AI RMF, Grafana Bloomberg

Talking points: Quick scan; share if audience wants to explore.

Timing: 15 sec

Key Concepts Reference

The Prompt (Exact)

“What metrics are important for evaluating the impact of AI on processing and data management workflows, and how does data quality play a role?”

Data Quality Is the Constraint

Analogy: Data quality is like the foundation of a building—AI can’t fix cracks in the foundation. Operational wins only matter if data remains fit for client use. “Measure, gate, ship” — not “ship and hope.”

Five-Block Taxonomy

Source: Wang & Strong (1996), ISO 25012, DAMA, DORA; synthesized in *prompt-a-deep-research-report.md*.

Block	The question	Decision	Example Metrics
Data quality	Is the data correct enough to ship?	Ship / hold	Accuracy, completeness, timeliness
Operational workflow	Is the pipeline running reliably?	Deploy / rollback	Lead time, MTTR, change failure rate
Human workflow	Did we reduce manual work safely?	Automate / keep manual	Hours saved, error rate delta
AI-specific	Did the AI add value without unreliability?	Promote / reject	Faithfulness (LLM); precision/recall (Traditional)
Governance & risk	Can we prove what happened?	Audit / remediate	Auditability, traceability

Block 4 Phrasing

Did the AI step add value without introducing unreliability?

LLM Evaluation Toolkit

- **RAGAS:** Faithfulness, answer relevancy, context recall/precision — Document Search, RAG
- **LLM-as-a-judge:** Subjective scoring via second LLM; MLflow GenAI; NSTM theme quality
- **Human-in-the-loop:** Human review/validation; ground truth for promotion; sample production for ongoing quality
- **Pydantic:** Schema validation; validation success rate; Event-Driven Feeds, NSTM metadata

Component-Aware Evaluation

Separate retrieval quality from generation quality. Otherwise you cannot tell whether regressions come from: Data, Retrieval, Prompts, Model.

Measurement Playbook

1. Define SLIs/SLOs
 2. Derive error budget
 3. Champion-challenger
 4. Canary
 5. Rollback triggers
 6. Decide upfront: primary, guardrail, rollback
-

Worked Examples Summary

Example	Type	Key Metrics	Rollback Trigger
Event-Driven Feeds LLM	LLM	Faithfulness, answer relevancy; precision/recall >0.9/0.85; timeliness <60s	Faithfulness drops by X points
Document Search RAG	LLM	Context recall >0.85; Faithfulness >0.9; Citation correctness 100%	Correctness < baseline
Anomaly Detection	Traditional ML	Precision >0.85; Recall >0.9; Time to detect <5 min	Precision <0.8 or recall <0.85

References & Links

Five-Block Taxonomy sources: - Wang & Strong (1996) — Beyond Accuracy: What Data Quality Means to Data Consumers - ISO/IEC 25012 — Data quality model - DORA metrics — Software delivery performance (lead time, change failure rate, MTTR) - prompt-a-deep-research-report.md — Bloomberg-ready taxonomy synthesis

Other: - Bloomberg Data - Bloomberg AI - Event-Driven Feeds - Data AI
Trainer Job - MLflow GenAI - RAGAS - Pydantic - Great Expectations -
OWASP LLM Top 10 - NIST AI RMF - Grafana Bloomberg

For full reference tables (Five-Block, Data Quality Dimensions, LLM evaluation toolkit, etc.) and glossary with learn-more links, see the main README.md in the muneeb directory.